

# AI-Based Driver Monitoring System

YOLO11s Object Detection + Transfer Learning

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01

# Problem, Context & Solution Overview

# Real-Time Driver Behavior Detection

Real-time computer vision monitors unsafe driver behaviors like phone use, smoking, missing seatbelts, and drowsiness to reduce distraction and crash risk.

## Visual Design

Highlight the slide with a dark navy background and bold title. Use cyan accents on the keywords "Driver Monitoring" and "YOLO11s" to draw attention. Keep layout clean with strong contrast to reinforce a modern, AI-powered safety solution.

## AI Solution

State the solution in one line: a YOLO11s-based computer vision system that analyzes live in-cabin camera input to detect unsafe driver behaviors in real time, enabling timely alerts and interventions while keeping text short and focused.

## Risk Behaviors

Explain behavior categories with crisp bullets: phone use, smoking, missing seatbelt, and drowsiness detected through eye-state changes. Link each behavior to distraction, slower reactions, and higher crash risk to make the safety impact immediately clear.

# Driver Safety

## Current Risk

### Phone Distraction

Drivers frequently handle phones for calls or messaging, pulling eyes and attention away from the road and sharply increasing collision risk.

### Smoking While Driving

Lighting, holding, or dropping cigarettes forces drivers to multitask, adding hand and visual distractions that delay reaction time.

### Seatbelt And Fatigue

Not wearing seatbelts worsens crash injuries, while drowsy drivers with heavy eyelids react slowly and may drift or microsleep.

VS

## AI Solution

### Camera Monitoring

An in vehicle or on device camera watches the driver zone, continuously capturing images or video frames without requiring manual input.

### YOLO11s Detection

YOLO11s identifies phones, seatbelts, cigarettes, and driver eye states in real time, supporting both still images and live streams.

### Alerts And Logging

The system triggers instant warnings and records each unsafe event, enabling rapid intervention and long term behavior analysis.

# End-to-End Detection Flow & Classes





02

# Dataset, Training Pipeline & Results

# From Raw Images to Deployed Model

01

## End-to-End Driver Monitoring Pipeline

Dataset to deployment: a clear, cyan-highlighted flow of six labeled, connected stages.



02

## Stepwise Process Overview

Dataset, audit, YOLO11s, transfer learning, evaluation, deployment shown with minimal text and directional arrows.





# Dataset Composition & Audit Findings

| Metric             | Count                                                |
|--------------------|------------------------------------------------------|
| Original images    | 9,884                                                |
| Duplicates removed | 156                                                  |
| Final images       | 9,728                                                |
| Train images       | 5,932                                                |
| Validation images  | 2,353                                                |
| Test images        | 1,443                                                |
| Classes            | 5 (Open Eye, Closed Eye, Cigarette, Phone, Seatbelt) |

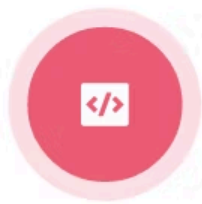
## Dataset Volume & Class Overview

Final cleaned dataset contains 9,728 RGB JPG images across five driver-state classes with bounding boxes.

## Split Summary & Data Audit Insights

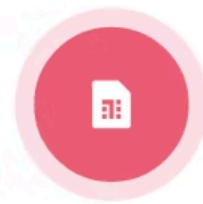
Train/validation/test splits: 5,932 / 2,353 / 1,443 images; class imbalance and minor eye-label noise identified.

# Training Configuration & Technical Setup



## Model & Training Setup

YOLO11s with transfer learning on custom dataset, optimized for deployable driver monitoring.



## Hyperparameters & Hardware

768×768 input, batch size 16, moderate epochs on Tesla T4 using pretrained weights for efficiency.

# Test Performance

## Global Metrics

| Average Value | YoY Growth |
|---------------|------------|
| 500           | 10%        |

On 1,443 unseen test images, the model achieves 90.46% precision and 85.80% recall, with mAP@0.5 at 91.01% and mAP@0.5:0.95 at 66.99%, indicating strong detection yet room for improvement.

## Quick Insights

| Average Value | YoY Growth |
|---------------|------------|
| 500           | 10%        |

High precision at 90.46% means most detections are correct, while 85.80% recall shows most true objects are captured; mAP@0.5 of 91.01% reflects robust localization, and 66.99% mAP@0.5:0.95 signals potential under stricter IoU.

# Class-Wise Highlights & Weaknesses

## Comparison Item A

### Strongest Class: Phone Usage

Phone achieves 98.25% mAP@0.5, delivering highly reliable detection of driver distraction behaviors.



## Comparison Item B

### Weakest Class: Cigarette Detection

Cigarette reaches 45.81% mAP@0.5:0.95; small size and stricter IoU thresholds significantly reduce accuracy.



03

# Demo, Failures, Future Work & Thanks

# From Detection to Monitoring Output



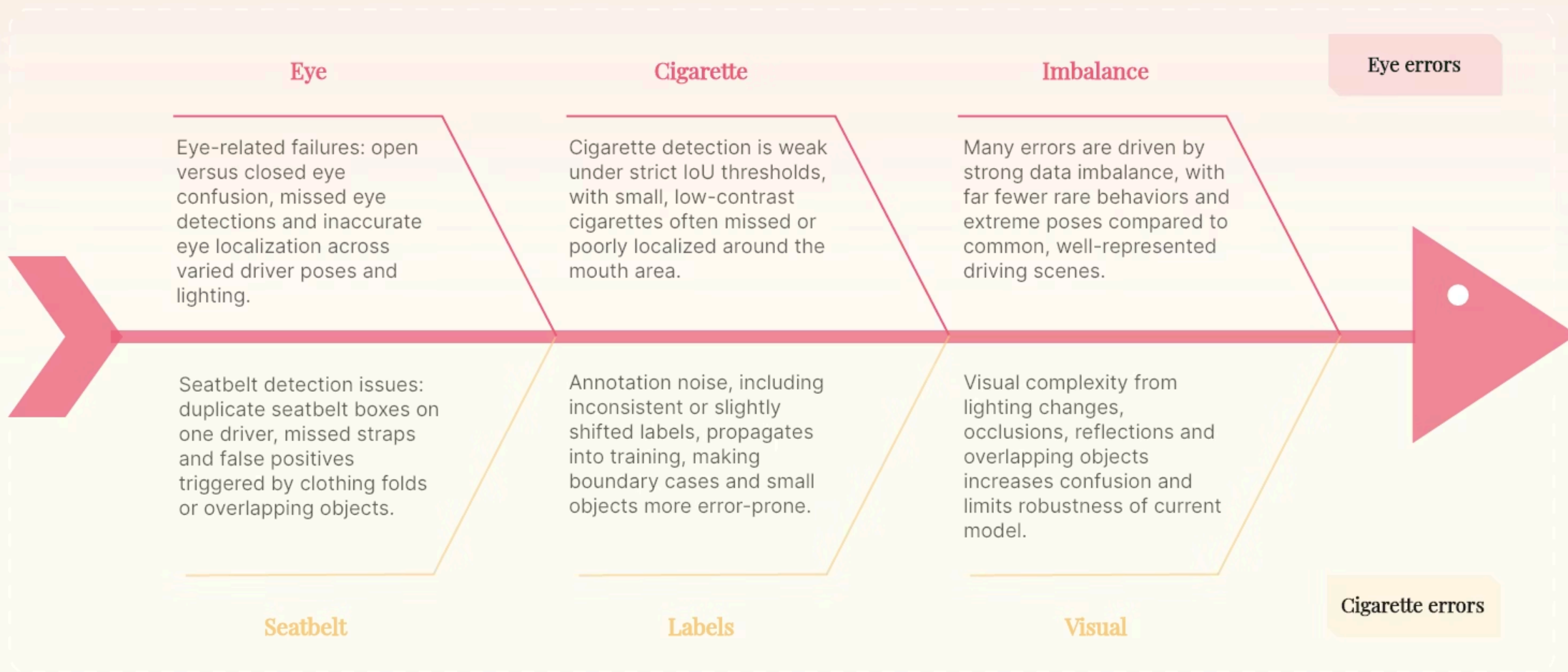
## End-to-end Monitoring Pipeline Snapshot

Deployment frame shows cyan boxes around driver, eyes, seatbelt, phone, cigarette, matching dark AI aesthetic.

## From Video Stream to Actionable Safety Signals

Flow: Video → YOLO11s → Detection → Labels, confidence → Alerts, behavior logs, inputs to safety coaching systems.

# Failure Cases & Limitations



Fishbone-style overview of eye, seatbelt and cigarette detection failures, rooted in data imbalance, noisy labels and visual complexity.



# Future Improvements & Closing Thanks



## Future Work & Model Enhancements

Improve class balance, annotation quality, and robustness with diverse data and targeted fine-tuning.



## Quote & Special Thanks

"A model detects objects. Great mentors reveal possibilities."  
Deep thanks to Eng. Amr Helal and Eng. Mohamed Waleed.





# Thanks

Expressing Gratitude and Appreciation in Everyday Life

Presenter: Rawan Adel